

數理統計

蘇南誠老師講義 · Quarto 教學整理版

蘇南誠

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1 首頁

數理統計講義

本講義由蘇南誠老師編授。感謝黃怡婷老師提供原始 講義檔案；本版保留原有公式、例題、圖形與英文統計術語，並加入一致的教學導覽架構。

1.1 使用方式

每章依序包含：

1. 學習目標
2. 必備的先備知識
3. 核心內容與原講義例題
4. 本章摘要
5. 常見錯誤
6. 章末練習

！ 編輯原則

目前版本以「忠實轉入原教材 + 建立一致骨架」為主。後續逐章修訂時，可補入中文導讀、完整證明、例題解答、課堂活動及難度標記。

1.2 課程主線

機率基礎 → 多變量分配 → 常用分配 → 抽樣推論 → 漸近理論 → 最大概似 → 充分性與完備性。

1.3 原始教材

Hogg, McKean, and Craig, *Introduction to Mathematical Statistics*, 2019。

2 機率與單變量分配

本章定位

本章保留原有英文術語、公式與例題，並整理為可連續閱讀的教學形式。

學生在大二已使用 Hogg, Tanis & Zimmerman 的 *Probability and Statistical Inference* 學過機率基礎，因此本章**不是重新教授機率論**，而是採取

Recall $\longrightarrow$ Deepen $\longrightarrow$ Connect

的三階段安排：先快速喚醒已學內容，再深化數理統計所需的推導工具，最後連接抽樣分配、收斂與後續推論。標題保留 Hogg, McKean & Craig Chapter 1 的 section numbering，方便對照課本；各節不採等比例講解。

2.1 學習目標

完成本章後，學生應能：

- 以樣本空間與事件描述隨機實驗
- 運用條件機率、全機率公式與 Bayes 定理
- 辨認離散與連續隨機變數的分配
- 計算期望、變異數、動差與動差生成函數
- 使用 Markov、Chebyshev 與 Jensen 不等式
- 由 Chebyshev 不等式推導樣本平均數的依機率收斂

2.2 必備的先備知識

集合符號、基本微積分與有限和。機率法則、條件機率、隨機變數及 pmf / pdf / cdf 視為大二已學內容，課堂中以診斷題與例題快速喚醒。

教學節奏

- **Stage 1 — Recall (快速)** : Sections 1.1–1.5，以 60–90 分鐘的 example-driven review 喚醒基本概念；忘記的內容在例題中補回，不重走完整的 Definition–Theorem–Proof 流程。
- **Stage 2 — Deepen (主體)** : Sections 1.6–1.9。1.6 與 1.7 的 pmf / pdf 基礎快速帶過，把時間放在 transformation；1.8 與 1.9 深入處理 $E[g(X)]$ 、moments、variance 與 MGF 的推導。
- **Stage 3 — Connect (橋接)** : Section 1.10。以重要不等式連接 sample mean、Chebyshev、convergence in probability 與後續 mathematical statistics。

Stage 1 — Recall：快速喚醒

學過的，快速喚醒；忘記的，在例題中補回來。

本階段以概念辨識與短例題為主。學生應能迅速回答 $P(A | B)$ 、 $F_X(x) = P(X \leq x)$ 、pmf/pdf 的差異及 cdf 的基本性質；細部證明可依診斷結果選講。

2.3 1.1 Introduction

Definition of experiment

- Random experiments
 - Outcome cannot be predicted with certainty.

Sample space

- Outcome (Sample) space
 - Is the collection of all possible outcomes.
 - Is known.
 - Is denoted as $\mathcal{S}$.

Event

- Event
 - Is a part of the collection of outcomes in $\mathcal{S}$.
 - Is denoted as A .
 - $A \subset \mathcal{S}$.

Frequency and relative frequency

- The probability of an event A is called the chance of A occurring.
- For example, a fair coin is tossed repeatedly n times.
- Count the number of times that event A (Head) occurs.
- The number is called frequency and is denoted as $N(A)$.
- The relative frequency is $N(A)/n$.
 - $n = 10$
 - H T H H T T H H T H
 - H: 6/10; T: 4/10

Features of relative frequency

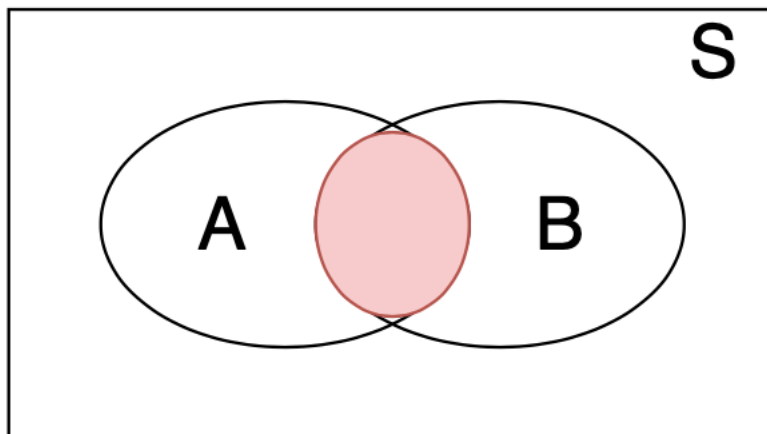
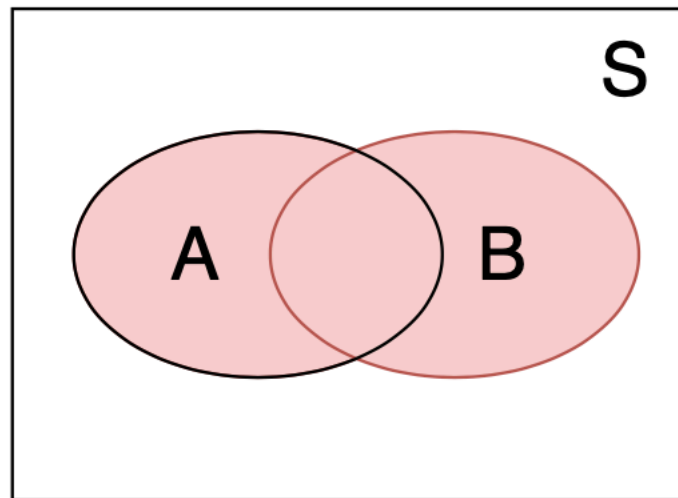
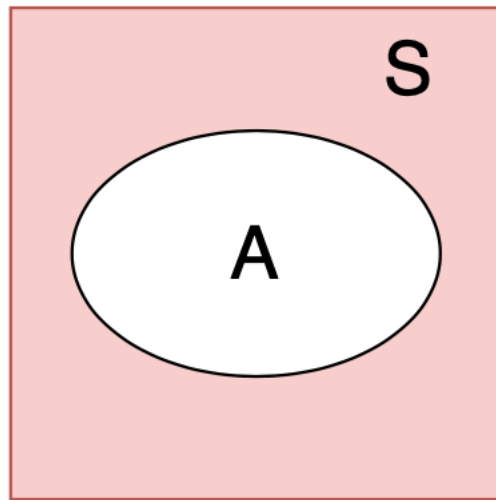
- As n tends to ∞ , relative frequency tends to be a fixed value.
- Suppose this fixed value is p .
- p is considered as the future performance of outcomes (event A) of the experiment.
- Tossing coin experiment n times
 - $S = \{H, T\}$
 - $N(S) = N(H) + N(T) = n$
 - $N(S)/n = 1$

2.4 1.2 Sets

Algebra of sets

- Null or empty set
 - $\emptyset$
- Subset
 - $A \subset B$
- Intersection
 - $A \cap B$
- Union
 - $A \cup B$
- Complement
 - A^C or A'

Venn Diagrams



Properties for set operations

- Commutative law

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

- Associative law

$$(A \cup B) \cup C = A \cup (B \cup C)$$
$$(A \cap B) \cap C = A \cap (B \cap C)$$

Properties for set operations

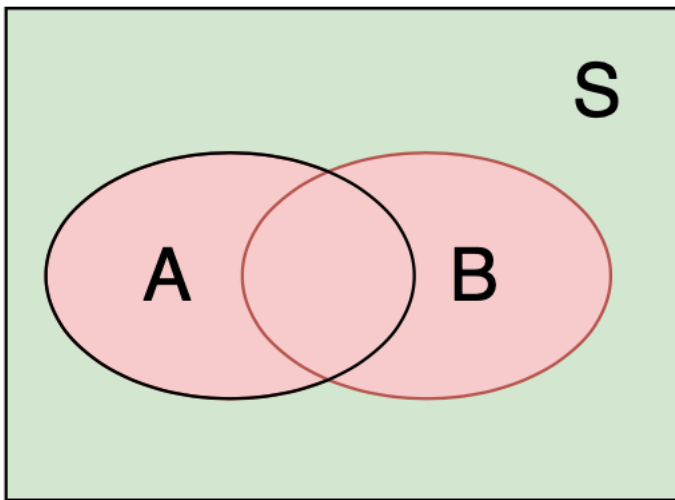
- Distributive law

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

- De Morgan's law

$$(A \cup B)' = A' \cap B'$$
$$(A \cap B)' = A' \cup B'$$

Venn Diagrams



More than 2 sets

- Let $A_1, A_2, \dots, A_n$ be any sets.
- Define

$$- A_1 \cup A_2 \cup \dots \cup A_n = \{x : x \in A_i, \text{ for some } i = 1, 2, \dots, n\} = \bigcup_{i=1}^n A_i.$$
$$- A_1 \cap A_2 \cap \dots \cap A_n = \{x : x \in A_i, \text{ for all } i = 1, 2, \dots, n\} = \bigcap_{i=1}^n A_i.$$

Countable sets

- Let $A_1, A_2, \dots, A_n, \dots$ be a sequence of sets.
- Define
 - $A_1 \cup A_2 \cup \dots \cup A_n \cup \dots = \{x : x \in A_i, \text{ for some } i = 1, 2, \dots, n, \dots\} = \bigcup_{n=1}^{\infty} A_n.$
 - $A_1 \cap A_2 \cap \dots \cap A_n \cap \dots = \{x : x \in A_i, \text{ for all } i = 1, 2, \dots, n, \dots\} = \bigcap_{n=1}^{\infty} A_n.$

Supplementary Example 1.2-A

- Let $\mathcal{S} = \{1, 2, 3, \dots\}.$
- Let $A_n = \{1, 3, \dots, 2n-1\}$ and $B_n = \{n, n+1, \dots\}$ for $n = 1, 2, 3, \dots$
- Find $\bigcup_{n=1}^{\infty} A_n, \bigcap_{n=1}^{\infty} A_n, \bigcup_{n=1}^{\infty} B_n, \bigcap_{n=1}^{\infty} B_n.$

Monotone (nondecreasing) sets

- Let $A_1, A_2, \dots, A_n, \dots$ be a sequence of nondecreasing sets, i.e.,

$$A_n \subset A_{n+1}, n = 1, 2, 3, \dots$$

- Define

$$\lim_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} A_n.$$

Monotone (nonincreasing) sets

- Let $A_1, A_2, \dots, A_n, \dots$ be a sequence of nonincreasing sets, i.e.,

$$A_n \supset A_{n+1}, n = 1, 2, 3, \dots$$

- Define

$$\lim_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} A_n.$$

Set functions

- Set function (集合函数)
 - Is a function that map sets into real numbers.

Common set functions

- Suppose $\mathcal{S}$ be a set of real number.
 - $Q(A) = \int_A f(x)dx$
 - $Q(A) = \int \int_A g(x, y)dxdy$
- Suppose $\mathcal{C}$ be a set of integers.

- $Q(A) = \sum_{x \in A} f(x)$
- $Q(A) = \sum \sum_{(x,y) \in A} g(x,y)$

Supplementary Example 1.2-B

- Suppose $\mathcal{S}$ be the interval of positive real numbers, i.e., $\mathcal{S} = (0, \infty)$.
- Let A be a subset of $\mathcal{S}$.
- Define
 - $Q(A) = \int_A e^{-x} dx$, provided the integral exists.
- Find $Q((1, 3))$.
- Find $Q((5, \infty))$.
- Find $Q((1, 3) \cup [3, 5))$.
- Find $Q(\mathcal{S})$.

2.5 1.3 Probability Set Function

Recall frequency

- The probability of an event A equals p .
- Denote as $P[A]$.
- $P[A]$ is a set function that associates A with the relative frequency.

Definition of Probability

定義 2.1.

- Let $\mathcal{S}$ be a sample space.
- Let $\mathcal{B}$ be the set of events.
- Let P be a real-valued set function defined on $\mathcal{B}$.
- P is a probability set function if it satisfies
 - $P[A] \geq 0$, for all $A \in \mathcal{B}$.
 - $P[\emptyset] = 0$
 - If $\{A_n\}$ is a sequence of events in $\mathcal{B}$ and $A_m \cap A_n = \emptyset$ for all $m \neq n$, then

$$P[A_1 \cup A_2 \cup A_3 \cup \dots] = \sum_{i=1}^{\infty} P[A_i]$$

Special terminology

- Mutually exclusive events
 - $A_i \cap A_j = \emptyset, \forall i \neq j$.
 - $A_1, \dots, A_k$ are disjoint sets.
 - The union is referred to as a disjoint union.
- Exhaustive events, if
 - $A_i \cap A_j = \emptyset, \forall i \neq j$.
 - $\cup_{n=1}^{\infty} A_n = \mathcal{S}$.

Special terminology

- Mutually exclusive and exhaustive events
 - $A_i \cap A_j = \emptyset, \forall i \neq j$.
 - $A_1, \dots, A_k$ are disjoint sets.
 - $A_1 \cup A_2 \cup \dots \cup A_k = \mathcal{S}$.
- It is a partition of $\mathcal{S}$.

Theorem

定理 2.1. For each event $A \in \mathcal{B}$,

$$P[A] = 1 - P[A']$$

Theorem

定理 2.2.

$$P[\emptyset] = 0$$

Theorem

定理 2.3. If events A and B are such that $A \subset B$, then

$$P[A] \leq P[B].$$

Theorem

定理 2.4. For each event $A \in \mathcal{B}$,

$$P[A] \leq 1.$$

Theorem

定理 2.5. If A and B are any two events in $\mathcal{S}$, then

$$P[A \cup B] = P[A] + P[B] - P[A \cap B].$$

Theorem

定理 2.6. If A , B and C are any three events, then

$$P[A \cup B \cup C] = P[A] + P[B] + P[C] - P[A \cap B] - P[A \cap C] - P[B \cap C] + P[A \cap B \cap C].$$

Example 1.3.1

- Consider one red die and one white die.
- Let the outcome be the ordered pair.
- Assume these two dice is repeatedly cast under the same condition.
- The cast of this pair of dice is a random experiment.
- This is a random experiment.
- Find the sample space $\mathcal{S}$.

Example 1.3.1 (continued)

- Assume the probability set function assigns a probability of $1/36$ to each of the 36 points in $\mathcal{S}$.
- If $C_1 = \{(1, 1), (2, 1), (3, 1), (4, 1), (5, 1)\}$ and $C_2 = \{(1, 2), (2, 2), (3, 2), (4, 2), (5, 2)\}$.
- Find $P[C_1]$, $P[C_2]$, $P[C_1 \cup C_2]$ and $P[C_1 \cap C_2]$.

Inclusion exclusion formula

- Consider a sequence of events $\{C_n, n = 1, 2, \dots\}$.
- Let $p_1 = P[C_1] + P[C_2] + P[C_3]$.
- Let $p_2 = P[C_1 \cap C_2] + P[C_1 \cap C_3] + P[C_2 \cap C_3]$.
- Let $p_3 = P[C_1 \cap C_2 \cap C_3]$.
- Then

$$P[C_1 \cup C_2 \cup \dots \cup C_k] = p_1 - p_2 + p_3 - \dots + (-1)^{n+1} p_n.$$

Bonferroni's inequality

- Consider two events C_1 and C_2 .
- We have

$$P[C_1 \cap C_2] \geq P[C_1] + P[C_2] - 1$$

Boole's inequality

定理 2.7.

- Let $\{C_n\}$ be an arbitrary sequence of events.
- Then

$$P\left[\bigcup_{n=1}^{\infty} C_n\right] \leq \sum_{n=1}^{\infty} P[C_n].$$

Equally likely principle

- Let $x_i, i = 1, \dots, m$, be all possible outcomes of an experiment.
- Suppose m is the total number of ways that the random experiment can terminate.
- If each of e_i has the same probability of occurring, m outcomes are equally likely.
- We have

$$P[e_i] = \frac{1}{m}, \quad i = 1, 2, \dots, m.$$

- Let $A \in \mathcal{S}$ and define

$$P[A] = \sum_{x_i \in A} \frac{1}{m} = \frac{\#(A)}{m}.$$

Example 1.3.2

- Toss a fair coin twice
- Possible outcomes: TT, HT, TH, HH
- Assume

$$P[TT] = P[HT] = P[TH] = P[HH] = \frac{1}{4}$$

- Consider $A = \{\text{at least one } H\}$

$$P[A] = P[HT + TH + HH] = P[HT] + P[TH] + P[HH] = \frac{3}{4}$$

Counting rule – Multiplication Principle

- Suppose an experiment E_1 has n_1 outcomes.
- Each of possible outcomes in E_1 , an experiment has n_2 outcomes
- The composite experiment $E_1 E_2$ consisting of performing E_1 and then E_2 has $n_1 n_2$ possible outcomes.

More on multiplication Principle

- Suppose experiment E_i has n_i outcomes, $i = 1, 2, \dots, m$.
- The composite experiment $E_1 E_2 \dots E_m$ has $n_1 n_2 \dots n_m$ possible outcomes.

Supplementary Example 1.3-A — Dinner choices

- Choices of dinner
 - E_1 : soup or tomato juice
 - E_2 : steak or shrimp
 - E_3 : French fried potatoes, mashed potatoes or a baked potatoes
 - E_4 : corn or pea
 - E_5 : jelly, tossed salad, cottage cheese or coleslaw
 - E_6 : cake, cookies, pudding, brownie, vanilla ice cream, chocolate ice cream or orange sherbet
 - E_7 : coffee, tea, milk or punch
- How many different dinner selections are possible?

Counting methods

- Permutation
- Combination
- Sampling with replacement
- Sampling without replacement
- Ordered sample
- Unordered sample

Common counting rules

- Let A be a set with n elements.
- A permutation of n objects equals $n!$.
- Suppose k -tuples whose components are elements of A , then there are n^k such k -tuples whose components are elements of A .
- Suppose $k < n$ and we are interest in k -tuples whose elements are distinct. Then, the number of k permutations taken from A equals

$$P_k^n = \frac{n!}{(n-k)!}.$$

Common counting rules

- Let A be a set with n elements.
- Suppose $k < n$ and we are interest in k -tuples whose elements are distinct.
- Suppose the order is not important.

- Then, the number of k permutations taken from A equals

$$C_k^n = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Remark

- $C_r^n = \binom{n}{r}$ is called binomial coefficients.
- The binomial expansion

$$(a+b)^n = \sum_{r=1}^n \binom{n}{r} b^r a^{n-r}$$

Definition

定義 2.2. Each of the

$$C_k^n = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

permutations of n objects, k of one type and $n - r$ of another type, is called a distinguishable permutation.

Remark

- Suppose there are n objects.
- n_1 are similar, n_2 are similar, ..., n_s are similar.
- $n = \sum_{i=1}^s n_i, 0 \leq n_i \leq n$.
- The number of distinguishable permutations of the n objects is

$$\binom{n}{n_1, n_2, \dots, n_s} = \frac{n!}{n_1! n_2! \dots n_s!}$$

- It is called multinomial coefficients.

Properties of probability

定理 2.8.

- Let $\{C_n\}$ be a nondecreasing sequence of events.
- Then

$$\lim_{n \rightarrow \infty} P[C_n] = P \left[\lim_{n \rightarrow \infty} C_n \right] = P \left[\bigcup_{n=1}^{\infty} C_n \right].$$

Properties of probability

定理 2.9.

- Let $\{C_n\}$ be a nonincreasing sequence of events.
- Then

$$\lim_{n \rightarrow \infty} P[C_n] = P\left[\lim_{n \rightarrow \infty} C_n\right] = P\left[\bigcap_{n=1}^{\infty} C_n\right].$$

2.6 1.4 Conditional Probability and Independence

Example 1.4.7

	Early (E)	Late (L)	Totals
Red (R)	5	8	13
Yellow (Y)	3	4	7
Totals	8	12	20

$$\bullet P[R|E] = \frac{5}{8} = \frac{N(R \cap E)}{N(E)} = \frac{N(R \cap E)/20}{N(E)/20} = \frac{P[R \cap E]}{P[E]}$$

Definition

定義 2.3. The conditional probability of an event A , given that event B has occurred, is defined by

$$P[A|B] = \frac{P[A \cap B]}{P[B]}$$

provided that $P[B] > 0$.

Properties

- $P[A|B] \geq 0$
- $P[B|B] = 1$
- $P[A \cap B] = P[B] \times P[A|B]$ (Multiplication rule)

Properties

- If $A_1, A_2, \dots$ are mutually exclusive events, then

$$P[\bigcup_{n=1}^{\infty} A_n | B] = \sum_{n=1}^{\infty} P[A_n | B]$$

Example 1.4.4

- Consider an ordinary deck of playing cards.
- Cards are to be drawn successively at random without replacement.
- Compute the probability that the third spade appears in the first six cards drawn.

Define

A = the event of two spades in the first five cards

B = the event of a spade on the sixth draw

Remark

- The probability that three events A , B and C , all occur is given by the multiplication rule,

$$P[A \cap B \cap C] = P[A]P[B|A]P[C|A \cap B]$$

Law of total probability

- Let $\{C_i, i = 1, 2, \dots, k\}$ be a partition of $\mathcal{S}$.
- Let C be an event.
- Then

$$P[C] = \sum_{i=1}^k P[C_i] \times P[C|C_i]$$

Bayes' theorem

- Let $\{C_i, i = 1, 2, \dots, k\}$ be a partition of $\mathcal{S}$.
- Let C be an event.
- Suppose $P[C] > 0$.
- Then

$$P[C_j|C] = \frac{P[C_j] \times P[C|C_j]}{\sum_{i=1}^k P[C_i] \times P[C|C_i]}.$$

Example 1.4.5

- The bowl A_1 contains three red and seven blue chips.
- The bowl A_2 contains eight red and two blue chips.
- All chips are identical in size and shape.
- A die is cast and bowl A_1 is selected if five or six spots show on the side that is up; otherwise, bowl A_2 is selected.
- Assign $P[A_1] = 2/6$ and $P[A_2] = 4/6$.
- The selected bowl is handed to another person and one chip is taken at random.

- Let B denote an event of a red chip is taken.
- Find $P[A_1|B]$ and $P[A_2|B]$.

Bayesian inference terminology

- The prior probability – the probability of $P[A_i]$.
- The posterior probability – the probability of $P[A_i|B]$.

Definition

定義 2.4. Events A and B are independent if and only if

$$P[A \cap B] = P[A]P[B]$$

Otherwise, A and B are called dependent events.

Remark

- Independent events are also called statistically independent, stochastically independent or independent in a probabilistic sense.
- The definition always holds If $P[A] = 0$ or $P[B] = 0$.

Example 1.4.8

- A red die and a white die are rolled.
- Let $A = \{4 \text{ on the red die}\}$.
- Let $B = \{\text{sum of dice is odd}\}$
- Show that A and B are independent events.

Example 1.4.9

- A red die and a white die are rolled.
- Let $A = \{5 \text{ on the red die}\}$.
- Let $B = \{\text{sum of dice is 11}\}$
- Show that A and B are dependent events.

Properties

- If A and B are independent events, then the following pairs of events are also independent:
 1. A and B'
 2. A' and B
 3. A' and B'

Supplementary Example 1.4-A

- An urn contains four balls numbered 1, 2, 3, and 4.
- One ball is to be drawn at random from the urn.
- Let $A = \{1, 2\}$, $B = \{1, 3\}$ and $C = \{1, 4\}$.
- Show that A and B , A and C , B and C are independent events, but not A , B and C .

Definition

定義 2.5. Events A , B and C are mutually independent if and only if the following two conditions hold:

1. A , B and C are pairwise independent; that is,

$$P[A \cap B] = P[A]P[B], P[A \cap C] = P[A]P[C], P[B \cap C] = P[B]P[C]$$

2. $P[A \cap B \cap C] = P[A]P[B]P[C]$

Properties

- If A , B and C are mutually independent events, then the following pairs of events are also independent:
 1. A and $B \cap C$
 2. A and $B \cup C$
 3. A' and $B \cap C'$
- A' , B' and C' are mutually independent.

Remark

- Many experiments consist of a sequence of n trials that are mutually independent.
- If the outcomes of the trials are not related to each other, then events, such that each is associated with a different trials, should be independent in the probability sense.
- If A_i is associated with the i th trials, $i = 1, 2, \dots, n$, then

$$P[A_1 \cap A_2 \cap \dots \cap A_n] = P[A_1]P[A_2] \dots P[A_n]$$

Example 1.4.11

- A compute system is built so that if component K_1 fails, it is bypassed and K_2 is used.
- If K_2 fails, then K_3 is used.
- Suppose that the probability that K_1 fails is 0.01, that K_2 fails is 0.03, and that K_3 fails is 0.08.

- Assume that the failures are mutually independent events.
- Find $P[\text{Failure of the system}]$.

2.7 1.5 Random Variables

Definition

定義 2.6.

- Consider a random experiment with a sample space $\mathcal{S}$.
- A function X assigns one and only one real number $X(c) = x$ to each element $c \in \mathcal{S}$.
- X is called a random variable.
- The space or range of X is the set of real numbers $\mathcal{D} = \{x : x = X(c), c \in \mathcal{S}\}$.

Types of random variables

- Discrete random variables – $\mathcal{D}$ is a countable set.
- Continuous random variables – $\mathcal{D}$ is an interval of real numbers.

Distribution of a random variable

- Given a random variable X , its range $\mathcal{D}$ becomes the sample space of interest.
- Besides inducing the sample space $\mathcal{D}$, X also induces a probability.
- We call the distribution of X .

Discrete distribution

- Let X be a discrete random variable with a finite space $\mathcal{D} = \{d_1, \dots, d_m\}$.
- Define

$$p_X(d_i) = P[\{c : X(c) = d_i\}], i = 1, 2, \dots, m.$$

which is called the probability mass function (pmf) of X .

- The probability of event D is

$$P_X(D) = \sum_{d_i \in D} p_X(d_i), D \subset \mathcal{D}.$$

Example 1.5.1

- Let X be the sum of the upfaces on a roll of a pair of fair 6-sided dice, each with the numbers 1 through 6 on it.
- The sample space is $\mathcal{S} = \{(i, j) : 1 \leq i, j \leq 6\}$.
- Find the pmf of X .

Example 1.5.1 (continued)

- Let B_1 denote the event of 7 and 11.
- Let B_2 denote the event of 2, 3 and 12.
- Find $P_X[B_1]$ and $P_X[B_2]$.

Continuous distribution

- Let X be a continuous random variable with an interval of real numbers as the space $\mathcal{D}$.
- For the continuous random variables, the simple events of interest are intervals.
- Determine a nonnegative function $f_X(x)$ such that for any interval of real numbers $(a, b) \in \mathcal{D}$, the induced probability distribution of X , $P_X(\cdot)$, is defined as

$$P_X[(a, b)] = P[\{c \in \mathcal{S} : a < X(c) < b\}] = \int_a^b f_X(x) dx.$$

which is the probability that X falls between a and b is the area under the curve $y = f_X(x)$ between a and b .

- The function f_X is formally defined as the probability density function (pdf) of X .

Example 1.5.2

- Let X denote the outcome when a point is selected at random from an interval $[0, 1]$.
- If the experiment is performed in a fair manner.
- Assume

$$P_X[(a, b)] = b - a, 0 < a < b < 1.$$

which is the length of the interval.

- The pdf of X is

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $P[\{X < \frac{1}{8}\} \cup \{X > \frac{7}{8}\}]$.

Example 1.5.2 (continued)

- For any point a , $0 < a < 1$.
- Choose n_0 such that $0 < a - n_0^{-1} < a < a + n_0^{-1} < 1$.
- We have

$$P[X = a] \leq P\left[a - \frac{1}{n} < X < a + \frac{1}{n}\right] = \frac{2}{n}, \forall n \geq n_0.$$

- Since $2/n \rightarrow 0$ as $n \rightarrow \infty$ and a is arbitrary, we conclude that $P(X = a) = 0$ for all $a \in (0, 1)$.

Cumulative distribution function

- Let X be a random variable.

- Define

$$\begin{aligned} F_X(x) &= P_X[(-\infty, x)] = P[X \leq x] \\ &= P[\{c \in \mathcal{S} : X(c) \leq x\}], \quad -\infty < x < \infty. \end{aligned}$$

the cumulative distribution function of X .

- Use cdf as an abbreviate.

Example 1.5.4

- Let X denote the outcome when a point is selected at random from an interval $[0, 1]$.
- If the experiment is performed in a fair manner.
- Assume

$$P_X[(a, b)] = b - a, \quad 0 < a < b < 1.$$

which is the length of the interval.

- The pdf of X is

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find cdf of X .

Example 1.5.4 (continued)

- $F_X(x)$ and $f_X(x)$ are related, i.e.,

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad \forall x \in R$$

and

$$\frac{d}{dx} F_X(x) = f_X(x), \quad \forall x \in R, \text{ except for } x = 0 \text{ and } x = 1.$$

Remark

- X and Y are equal in distribution and write $X \stackrel{D}{=} Y$ if and only if $F_X(x) = F_Y(x)$, for all $x \in R$.

Supplementary Example 1.5-A

- Let X denote the outcome when a point is selected at random from an interval $[0, 1]$.
- Define the random variable Y as $Y = 1 - X$.
- $Y \neq X$.
- The space of Y is the interval $(0, 1)$, the same as X .

- The cdf of Y is

$$F_Y(y) = \begin{cases} 0 & y < 1 \\ P[Y \leq y] = 1 - (1 - y) = y & 0 \leq y < 1 \\ 1 & y \geq 1 \end{cases}.$$

- Y has the same cdf as X , i.e., $X \stackrel{D}{=} Y$, but $Y \neq X$.

Properties of cdf

定理 2.10.

- Let X be a random variable with cumulative distribution function $F(x)$.
- Then
 - For all a and b , if $a < b$, then $F(a) \leq F(b)$ (F is nondecreasing).
 - $\lim_{x \rightarrow -\infty} F(x) = 0$
 - $\lim_{x \rightarrow \infty} F(x) = 1$
 - $\lim_{x \rightarrow x_0-} F(x) = F(x_0)$ (F is right continuous).

Properties of cdf

定理 2.11.

- Let X be a random variable with cumulative distribution function $F(x)$.
- Then, for $a < b$,

$$P[a < X \leq b] = F_X(b) - F_X(a).$$

Properties of cdf

定理 2.12.

- Let X be a random variable with cumulative distribution function $F(x)$.
- Then,

$$P[X = x] = F_X(x) - F_X(x-),$$

for all $x \in \mathbb{R}$, where $F_X(x-) = \lim_{z \rightarrow x-} F_X(z)$.

Example 1.5.5

- Let X be the lifetime in years of a mechanical part.
- Assume that X has the cdf

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x} & 0 \leq x \end{cases}.$$

- Find the pdf of X and $P[a < X \leq 3]$.

Properties of pdf or pmf

- The total probability associated with a random variable X of the discrete type with pmf $p_X(x)$ or of the continuous type with pdf $f_X(x)$ is 1.
- It must be true that

$$\sum_{x \in \mathcal{D}} p_X(x) = 1 \quad \text{and} \quad \int_{\mathcal{D}} f_X(x) dx = 1,$$

where $\mathcal{D}$ is the space of X .

Example 1.5.8

- Let X has the pdf

$$f_X(x) = \begin{cases} cx^3 & 0 < x < 2 \\ 0 & \text{elsewhere,} \end{cases}$$

for an appropriate constant c .

- Find c .

Stage 2 — Deepen : 從會算到會推導

本階段開始放慢。Sections 1.6–1.7 的前段定義是 Recall 與 Deepen 的橋接，課堂重點不是再次背誦 pmf / pdf，而是從

$$X \longrightarrow g(X) \longrightarrow \text{distribution of } g(X)$$

進一步走向

$$X_1, \dots, X_n \longrightarrow T(X_1, \dots, X_n),$$

為 sampling distribution 建立工具。接著以 Sections 1.8–1.9 深化 expectation、moments、variance 與 MGF。

2.8 1.6 Discrete Random Variables

Discrete random variables

定義 2.7. A random variable is a discrete random variable if its space is either finite or countable.

Probability mass function

定義 2.8.

- Let X be a discrete random variable with space $\mathcal{D}$.
- The probability mass function (pmf) of X given by

$$p_X(x) = P[X = x], \text{ for } x \in \mathcal{D}.$$

Properties of Probability mass function

1. $p_X(x) > 0, x \in \mathcal{D}$.
2. $\sum_{x \in \mathcal{D}} p_X(x) = 1$.
3. $P[X \in A] = \sum_{x \in A} p_X(x)$, where $A \subset \mathcal{D}$.

Support

- Usually define $p_X(x) = 0$ if $x \notin S$.
- The domain of $p_X(x)$ is the set of R .
- To distinguish, the **support** of X is the set of outcomes with positive probabilities.

Hypergeometric distribution

- Consider a collection of $N = N_1 + N_2$ similar objects.
 - N_1 objects belong to the first class (red chip).
 - N_2 objects belong to the second class (blue chip).
- n objects are randomly selected from N objects.
- Let X be the number of objects drawing from the first class.
- Find the pmf of X .

Supplementary Example 1.6-A — Hypergeometric distribution

- There are 50 fishes in a small pond.
- 10 of which have been tagged.
- Suppose 7 fishes are caught randomly without replacement.
- Find the probability that exactly 2 tagged fishes are caught.

Transformation

- X has a pmf $f(x)$.
- Let $Y = u(X)$ for a given function $u(\cdot)$.
- Find the pmf of Y

Example 1.6.1

- Consider a sequence of independent flips of a coin, each resulting in a head (H) or a tail (T).
- Let X be the flip number on which the first head appear.
- Let Y be the number of flips before the first head, i.e., $Y = X - 1$.

- Find the pmf of Y .

Example 1.6.3

- Consider a sequence of independent flops of a coin, each resulting in a head (H) or a tail (T).
- Let X be the flip number on which the first head appear.
- If the first head appears on an odd number of flips, we pay the house one dollar, while if it appears on an even number of flips, we win one dollar from the house.
- Let Y denote our net gain.
- The space of Y is $-1, 1$
- Find the pmf of Y .

2.9 1.7 Continuous Random Variables

Continuous random variables

定義 2.9. A random variable is a continuous random variable if the cdf $F_X(x)$ is a continuous function for all $x \in \mathbb{R}$.

Probability density function

- Most continuous random variables are absolutely continuous; that is,

$$F_X(x) = \int_{-\infty}^x f_X(t) dt,$$

for some function $f_X(t)$.

- The function $f_X(t)$ is called a probability density function (pdf) of X .
- If $f_X(t)$ is also continuous, then the Fundamental Theorem of Calculus implies that

$$\frac{d}{dx} F_X(x) = f_X(x).$$

Support

- The support of a continuous random variable X consists of all points x such that $f_X(x) > 0$.
- Denote the support of X by S .

Properties of pdf

- The pdf $f(x)$ of X of the continuous type with space S .
- S is an interval or union of intervals.

- $f(x)$ satisfies
 - $f(x) \geq 0, x \in S$.
 - $\int_S f(x)dx = 1$.
 - If $(a, b) \subseteq S$, then

$$P[a < X < b] = \int_a^b f(x)dx.$$

Uniform distribution

- X has a uniform distribution if

$$f(x) = \frac{1}{b-a}, a \leq x \leq b$$

and 0 elsewhere.

- Denote $U(a, b)$.

Example 1.7.1

- Suppose a point is selected at random in the interior of a circle of radius 1.
- Let X be the distance of the selected point from the origin.
- The sample space for the experiment is

$$\mathcal{S} = \{(w, y) : w^2 + y^2 < 1\}.$$

- Assume subsets of $\mathcal{S}$ which have equal area are equilikely.
- Hence, the probability of the selected point lying in a set $A \subset \mathcal{S}$ is proportional to the area of A ; i.e.,

$$P[A] = \frac{\text{area of } A}{\pi}$$

- Find cdf and pdf of X .

Quantiles

定義 2.10.

- Let $0 < p < 1$.
- The quantile of order p of the distribution of a random variable X is a value ξ_p such that $P(X < \xi_p) \leq p$ and $P(X \leq \xi_p) \geq p$.
- It is also known as the $(100p)$ th percentile of X .

Quartiles

定義 2.11.

- The quantile of order 0.25, 0.5, 0.75 of the distribution of a random variable X is called the first quartile (q_1), median and the third quartile (q_3).
- The difference $IQ = q_3 - q_1$ is called the interquartile range of X .

Note

- Quantiles need not be unique even for continuous random variables with pdfs.
- Any point in the interval (2,3) serves as a median for the following pdf:

$$f_X(x) = \begin{cases} 3(1-x)(x-2) & 1 < x < 2 \\ 3(3-x)(x-4) & 3 < x < 4 \\ 0 & \text{elsewhere} \end{cases}.$$

Note

- A quantile, say ξ_p , is in the support of an absolutely continuous random variable X with cdf $F_X(x)$ then ξ_p is the unique solution to the equation:

$$\xi_p = F_X^{-1}(p),$$

where $F_X^{-1}(u)$ is the inverse function of $F_X(x)$.

Distribution function technique

- Let X be a random variable of the continuous type.
- Let $Y = u(X)$, where $u(\cdot)$ is a known function.
- Then the pdf of Y can be derived from

$$G(y) = P[Y \leq y] = P[u(X) \leq y]$$

Supplementary Example 1.7-A — Log gamma distribution

- Let X have a gamma distribution with pdf

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, 0 < x < \infty,$$

where $\alpha > 0$ and $\theta > 0$.

- Let $Y = \ln(X)$.
- Find the pdf of Y .

Theorem

定理 2.13.

- Let X be a random variable of the continuous type with pdf $f(x)$ with support S_X .
- Let $Y = g(X)$, where $g(\cdot)$ is a one-to-one differentiable function on S_X .
- Assume the inverse function g by $x = g^{-1}(y)$ and let

$$J = \frac{dx}{dy} = \frac{dg^{-1}(y)}{dy} (\text{Jacobian})$$

- Then the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|, \text{ for } y \in S_Y,$$

where S_Y is the support of Y and $S_Y = \{y = g(x) : x \in S_X\}$.

Supplementary Example 1.7-A (continued)

- Let X have a gamma distribution with pdf

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, 0 < x < \infty,$$

where $\alpha > 0$ and $\theta > 0$.

- Let $Y = \ln(X)$.
- Find the pdf of Y .

Example 1.5.6 — Mixture type of distribution

- Let X have cdf

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x+1}{2} & 0 \leq x < 1 \\ 1 & 1 \leq x \end{cases}.$$

- Find $P[-3 < X \leq \frac{1}{2}]$ and $P[X = 0]$.

2.10 1.8 Expectation of a Random Variable

Expectation – Continuous

定義 2.12.

- Let X be a random variable.

- If X with a continuous random variable with pdf $f(x)$ and

$$\int_{-\infty}^{\infty} |x|f(x)dx < \infty$$

then the expectation of X is

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx.$$

Expectation – Discrete

定義 2.13.

- Let X be a random variable.
- If X with a pmf $p(x)$ and

$$\sum_{x \in S} |x|p(x)$$

then the expectation of X is

$$E[X] = \sum_{x \in S} xp(x).$$

Note

- The expectation $E(X)$ is called the mathematical expectation of X , the expected value of X , or the mean of X .
- When the mean designation is used, $E(X)$ is denoted by μ ; i.e., $\mu = E(X)$.

Supplementary Example 1.8-A

- Let X have a pmf

$$f(x) = \frac{1}{3}, x \in S_X,$$

where $S_X = \{-1, 0, 1\}$.

- Find $E[X]$.

Example 1.8.3

- Let X have a pdf

$$f(x) = \begin{cases} 4x^3 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $E[X]$.

Expectation of a function of random variable (continuous)

定理 2.14.

- Let X be a random variable and let $Y = g(X)$ for some function g .
- If X with a continuous random variable with pdf $f_X(x)$ and

$$\int_{-\infty}^{\infty} |g(x)| f_X(x) dx < \infty$$

then the expectation of X is

$$E[Y] = \int_{-\infty}^{\infty} g(x) f(x) dx.$$

Expectation of a function of random variable (discrete)

定理 2.15.

- Let X be a random variable and let $Y = g(X)$ for some function g .
- If X with a pmf $p(x)$ and

$$\sum_{x \in S_X} |g(x)| p(x)$$

then the expectation of X is

$$E[Y] = \sum_{x \in S_X} g(x) p(x).$$

Supplementary Example 1.8-B

- Consider a sequence of independent flips of a coin, each resulting in a head (H) or a tail (T).
- Let X was the flip number on which the first head appeared.
- Let Y be the number of flips before the first head.
- Let $Z = e^{-Y}$.
- Find $E[Z]$.

Supplementary Example 1.8-C

- Let X be a continuous random variable with the pdf $f(x) = 2x$ which has support on the interval $(0, 1)$.
- Let $Y = 1/(1 + X)$.
- Find $E[Y]$.

Theorem

定理 2.16. When the expectation exists, it satisfies the following properties:

- If c is a constant, then $E[c] = c$.
- If c is a constant and $u(\cdot)$ is a function, then

$$E[cu(X)] = cE[u(X)].$$

- If c_1 and c_2 are constants and u_1 and u_2 are functions, then

$$E[c_1u_1(X) + c_2u_2(X)] = c_1E[u_1(X)] + c_2E[u_2(X)].$$

Comment

- If $c_1, \dots$ and c_k are constants and $u_1, u_2, \dots$ and u_k are functions, then

$$E\left[\sum_{i=1}^k c_i u_i(X)\right] = \sum_{i=1}^k c_i E[u_i(X)].$$

Example 1.8.4

- Let X have the pdf

$$f(x) = \begin{cases} 2(1-x) & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $E[X]$, $E[X^2]$ and $E[6X + 3X^2]$.

Example 1.8.5

- Let X have the pmf

$$f(x) = \begin{cases} \frac{x}{6} & x = 1, 2, 3 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $E[6X^3 + X]$.

Example 1.8.6

- Let us divide, at random, a horizontal line segment of length 5 into two parts.
- If X is the length of the left-hand part, it is reasonable to assume X has the pdf

$$f(x) = \begin{cases} \frac{1}{5} & 0 < x < 5 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $E[X(5 - X)]$.

2.11 1.9 Some Special Expectations

Settings

- Let X be a discrete random variable.
- Let the support of X be denoted as $S_X = \{u_1, u_2, \dots, u_k\}$.
- Assume

$$P[X = u_i] = f(u_i) = p_i, i = 1, 2, \dots, k$$

and

$$\sum_{i=1}^k p_i = 1.$$

Definition of mean

定義 2.14.

- Let X be a random variable whose expectation exists.
- The mean value μ of X is defined to be $\mu = E(X)$.

Definition of variance

定義 2.15.

- Let X be a random variable with finite mean μ and such that $E[(X - \mu)^2] < \infty$.
- The variance of X is defined to be $E[(X - \mu)^2]$.
- It is usually denoted by σ^2 or by $\text{Var}(X)$.

Supplementary Example 1.8-D

- Consider a discrete random variable X has a support $S_X = \{1, 2, 3\}$.
- Let $f(1) = \frac{3}{6}, f(2) = \frac{2}{6}, f(3) = \frac{1}{6}$.
- Find $E[X]$ and $\text{Var}(X)$.

Comment

- Show that

$$\text{Var}(X) = E[X^2] - \mu^2,$$

- The standard deviation of X is the positive square root of variance and is denoted as σ .

Class Example 1.9-A

- Let a and b be constants.
- Let $Y = aX + b$.
- Find $E[Y]$ and $\text{Var}(Y)$.

Example 1.9.1

- Suppose the random variable X has a uniform distribution with pdf

$$f_X(x) = 1/(2a), -a < x < a,$$

zero elsewhere.

- Find μ and σ^2 .

Example 1.9.2

- Let X has the pdf

$$f(x) = \begin{cases} \frac{1}{2}(x+1) & -1 < x < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find μ and σ^2 .

Example 1.9.5

- Let X has the pdf

$$f(x) = \begin{cases} \frac{1}{x^2} & 1 < x < \infty \\ 0 & \text{elsewhere} \end{cases}.$$

- Find μ .

Moment

- The r th moment is defined as

$$E[X^r] = \sum_{x \in S} x^r f(x).$$

- The r th moment about b is defined as

$$E[(X - b)^r] = \sum_{x \in S} (x - b)^r f(x).$$

- The r th factorial moment is defined as

$$E[(X)_r] = E[X(X - 1) \cdots (X - r + 1)].$$

Supplementary Example 1.9-A — Hypergeometric distribution

- Let X have a hypergeometric distribution where n objects are selected from $N = N_1 + N_2$.
- Find $\text{Var}(X)$.

Moment generating function

定義 2.16.

- Let X be a random variable.
- If there is a $h > 0$ such that

$$E[e^{tX}] = \sum_{x \in S} e^{tx} f(x)$$

exists and is finite for $-h < t < h$, then the function

$$M(t) = E[e^{tX}]$$

is called the moment-generating function (mgf) of X .

Comments

- If $t = 0$, $M(0) = 1$.
- If $S = \{b_1, b_2, \dots\}$, the mgf is

$$M(t) = e^{tb_1} f(b_1) + e^{tb_2} f(b_2) + e^{tb_3} f(b_3) + \dots,$$

- The coefficient of e^{tb_i} is

$$f(b_i) = P[X = b_i]$$

Comments

- Consider two random variables had two pmfs $f(x)$ and $g(y)$.
- Suppose the same space $S = \{b_1, b_2, \dots\}$ is defined.
- If

$$\begin{aligned} e^{tb_1} f(b_1) + e^{tb_2} f(b_2) + e^{tb_3} f(b_3) + \dots \\ = e^{tb_1} g(b_1) + e^{tb_2} g(b_2) + e^{tb_3} g(b_3) + \dots, \end{aligned}$$

for all t , $-h < t < h$, then

$$f(b_i) = g(b_i), \forall i$$

Example 1.9.3

- Suppose we have a fair spinner with the numbers 1, 2, and 3 on it.
- Let X be the number of spins until the first 3 occurs.

- Assuming that the spins are independent, the pmf of X is

$$p(x) = \frac{1}{3} \left(\frac{2}{3}\right)^{x-1}, x = 1, 2, 3, \dots$$

- Find the mgf of X .

Equivalence

定理 2.17.

- Let X and Y be random variables with moment generating functions $M_X(\cdot)$ and $M_Y(\cdot)$, respectively, existing in open intervals about 0.
- Then $F_X(z) = F_Y(z)$ for all $z \in R$ if and only if $M_X(t) = M_Y(t)$ for all $t \in (-h, h)$ for some $h > 0$.

Supplementary Example 1.9-B

- Suppose X has the mgf

$$M(t) = e^t \left(\frac{1}{10}\right) + e^{2t} \left(\frac{2}{10}\right) + e^{3t} \left(\frac{3}{10}\right) + e^{4t} \left(\frac{4}{10}\right), -\infty < t < \infty$$

- Find the pmf of X .

Class Example 1.9-B

- Show that $M'(0) = \mu$ and $M''(0) - [M'(0)]^2 = \sigma^2$.

Note

- If m is a positive integer and $M^{(m)}(t)$ denotes the m th derivative of $M(t)$.
- Then $M^{(m)}(0) = E[X^m]$.

Stage 3 — Connect : 由機率工具走向數理統計

本階段不把 inequalities 當作孤立公式。Markov、Chebyshev 與 Jensen 分別提供尾端機率控制、集中現象及凸性比較；其中 Chebyshev 不等式會直接連接樣本平均數與依機率收斂，為後續抽樣分配、漸近理論與統計推論鋪路。

2.12 1.10 Important Inequalities

Existence of moments

- Let X be a random variable and let m be a positive integer.

- Suppose $E[X^m]$ exists.
- If k is a positive integer and $k \leq m$, then $E[X^k]$ exists.

Markov's Inequality

定理 2.18.

- Let $u(X)$ be a nonnegative function of the random variable X .
- If $E[u(X)] < \infty$, random variable and let m be a positive integer.
- For every positive constant c ,

$$P[u(X) \geq c] \leq \frac{E[u(X)]}{c}.$$

Chebyshev's Inequality

定理 2.19.

- Let X be a random variable with finite variance σ^2 .
- For every positive constant k ,

$$P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}.$$

Example 1.10.1

- Let X has the pdf

$$f(x) = \begin{cases} \frac{1}{2\sqrt{3}} & -\sqrt{3} < x < \sqrt{3} \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $P[|X - \mu| \geq k\sigma]$ and the upper bound.

Example 1.10.2

- Let the random variable X of the discrete type have probabilities $1/8, 6/8, 1/8$ at the points $x = -1, 0, 1$, respectively.
- Find $P[|X - \mu| \geq k\sigma]$ and the upper bound.

Connect: sample mean and Chebyshev

令 $X_1, \dots, X_n$ 為獨立同分配 (iid) 的隨機變數，且

$$E(X_i) = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

樣本平均數為

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

由期望值與獨立性，

$$E(\bar{X}_n) = \mu, \quad \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

因此對任意 $\varepsilon > 0$ ，Chebyshev 不等式給出

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

這正是

$$\bar{X}_n \xrightarrow{P} \mu,$$

也就是樣本平均數依機率收斂到母體平均數；這個結果是弱大數法則的一個版本。

💡 往後連接

這裡的邏輯鏈是

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \implies \text{Chebyshev bound} \rightarrow 0 \implies \bar{X}_n \xrightarrow{P} \mu.$$

後續將把 $T(X_1, \dots, X_n)$ 視為 statistic，研究其 sampling distribution，並在漸近理論中系統整理 convergence in probability。參見 [Sampling and Inference](#) 與 [Asymptotics](#)。

Convex function

定義 2.17.

- A function ϕ defined on an interval (a, b) , $-\infty < a < b < \infty$, is said to be a convex function if for all x, y in (a, b) and for all $0 < \gamma < 1$,

$$\phi(\gamma x + (1 - \gamma)y) \leq \gamma\phi(x) + (1 - \gamma)\phi(y).$$

We say ϕ is strictly convex if the above inequality is strict.

Properties – Convex function

定理 2.20.

- If ϕ is differentiable on (a, b) , then
 - ϕ is convex if and only if $\phi'(x) \leq \phi'(y)$, for all $a < x < y < b$.

- ϕ is strictly convex if and only if $\phi'(x) < \phi'(y)$, for all $a < x < y < b$.
- If ϕ is twice differentiable on (a, b) , then
 - ϕ is convex if and only if $\phi''(x) \geq 0$, for all $a < x < b$.
 - ϕ is strictly convex if and only if $\phi''(x) > 0$, for all $a < x < b$.

Jensen's Inequality

定理 2.21.

- If ϕ is convex on an open interval I and X is a random variable whose support is contained in I and has finite expectation, then

$$\phi(E[X]) \leq E[\phi(X)]$$

- If ϕ is strictly convex, then the inequality is strict unless X is a constant random variable.

Example 1.10.3

- Let X be a nondegenerate random variable with mean μ and a finite second moment.
- Show that $(E[X])^2 \leq E[X^2]$.

Example 1.10.4

- Let $\{a_1, \dots, a_n\}$ be a set of positive numbers.
- Create a distribution for a random variable X by placing weight $1/n$ on each of the numbers $a_1, \dots, a_n$.
- The arithmetic mean (AM) equals

$$E[X] = \frac{1}{n} \sum_{i=1}^n a_i.$$

- The geometric mean (GM) equals $(a_1 a_2 \cdots a_n)^{1/n}$.
- The harmonic mean (HM) equals

$$\frac{1}{\frac{1}{n} \sum_{i=1}^n \frac{1}{a_i}}$$

- Show that $HM \leq GM \leq AM$.

2.13 本章摘要

本章依 **Recall** $\rightarrow$ **Deepen** $\rightarrow$ **Connect** 建立後續數理統計所需的共同語言。Stage 1 快速喚醒機率模型、條件機率、隨機變數與分配；Stage 2 深化 transformation、 $E[g(X)]$ 、moments、variance 與 MGF；Stage 3 以重要不等式連接樣本平均數、依機率收斂及後續推論理論。

2.14 常見錯誤

- 把事件與隨機變數混為一談
- 使用密度值代替區間機率
- 套用條件機率時忘記分母事件
- 引用不等式時沒有檢查非負性或凸性條件

2.15 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。
3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。
4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

授課提醒

本章保留完整材料供學生查閱，但「保留」不等於「逐頁等速講授」。Stage 1 可按班級診斷結果選題，Stage 2 應保留推導時間，Stage 3 則務必讓學生說明 Chebyshev 如何導出 $\overline{X}_n \xrightarrow{P} \mu$ 。

3 多變量分配與變數轉換

本章定位

本講義由蘇南誠老師編授。感謝黃怡婷老師提供原始數理統計講義檔案；本章保留原有英文術語、公式與例題，並整理為可連續閱讀的教學形式。

學生在大二已使用 Hogg, Tanis & Zimmerman 的 *Probability and Statistical Inference* 學過雙變量分配。該書 Chapter 4 已涵蓋 joint、marginal 與 conditional distributions、independence、covariance 與 correlation，Chapter 5.2 亦已介紹雙變量變數轉換。因此本章不將所有內容等量重教，而採取

Recall → Deepen → Connect

的三階段安排：先快速喚醒二維隨機變數的基本工具，再推廣到 n 維隨機向量、共變異矩陣與多維變數轉換，最後連接線性組合、random sample、sample mean 與 sample variance。標題依 Hogg, McKean & Craig Chapter 2 的內容順序安排，各節不採等比例講解。

3.1 學習目標

完成本章後，學生應能：

- 由聯合分配求邊際與條件分配
- 處理二維與多維隨機變數轉換
- 計算共變異數、相關係數與共變異矩陣
- 判斷隨機變數的獨立性
- 使用向量與矩陣表示多維隨機變數的平均與共變異結構
- 計算隨機變數線性組合的期望、變異數與共變異數
- 說明 random sample、sample mean 與 sample variance 如何連接後續統計推論

3.2 必備的先備知識

第 1 章的分配函數、期望與單變量變數轉換，以及偏微分與基本矩陣運算。joint pmf / pdf、marginal distribution、conditional distribution、independence、covariance、correlation 與二維 Jacobian 視為大二已學內容，課堂中以診斷題與代表性例題快速喚醒。

教學節奏

- **Stage 1 — Recall (快速)**：Sections 2.1–2.5，對照大二課本 Chapter 4 與 Section 5.2，以 60–90 分鐘的 example-driven review 喚醒 joint、marginal、conditional、independence、covariance、correlation 與二維 transformation；不重走完整的 Definition–

Theorem–Proof 流程。

- **Stage 2 — Deepen (主體)**: Sections 2.6–2.7。把二維概念推廣到 n 維 random vector，深入處理 covariance matrix、線性代數表示、多維 Jacobian，以及 one-to-one / many-to-one transformations。
- **Stage 3 — Connect (橋接)**: Section 2.8。以線性組合、random sample、 $\bar{X}$ 與 S^2 連接後續的 sampling distributions、sufficiency、likelihood 與 statistical inference。

Stage 1 — Recall：二維基本款快速喚醒

大二學過的，先確認能辨認、能設支持集、能完成積分；忘記的，再由代表性例題補回。

本階段的重點不是重複所有計算，而是確認學生能在離散與連續情形中，從 joint distribution 正確取得 marginal 與 conditional distributions，判斷 independence，計算 covariance / correlation，並在變數轉換時同時處理 Jacobian 與轉換後的支持集。

3.3 Distributions of two random variables

Definition

定義 3.1.

- Consider a random experiment with a sample space $\mathcal{C}$.
- Define consider two random variables X_1 and X_2 .
- Assign to each element c of $\mathcal{C}$ one and only one ordered pair of numbers $X_1(c) = x_1$, $X_2(c) = x_2$.
- Then,
 - (X_1, X_2) is a random vector.
 - The space of (X_1, X_2) is the set of ordered pairs

$$\mathcal{D} = \{(x_1, x_2) : x_1 = X_1(c), x_2 = X_2(c), c \in \mathcal{C}\}.$$

Cumulative distribution function (cdf)

- Let $A \subset \mathcal{D}$.
- Denote the probability of the event A as $P_{X_1, X_2}[A]$.
- $P_{X_1, X_2}[A]$ is uniquely determined by the cumulative distribution function (cdf), which is given by

$$F_{X_1, X_2}(x_1, x_2) = P[\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\}],$$

for all $(x_1, x_2) \in \mathbb{R}^2$.

Properties of cdf

- For all $a_1 < b_1$ and $a_2 < b_2$, show that

$$P[a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2] = F_{X_1, X_2}(b_1, b_2) - F_{X_1, X_2}(a_1, b_2) - F_{X_1, X_2}(b_1, a_2) + F_{X_1, X_2}(a_1, a_2)$$

Discrete random vector

定義 3.2.

- Suppose $\mathcal{D}$ is finite or countable.
- Let S denote the two-dimensional space of X_1 and X_2 .
- The joint probability mass function of X_1 and X_2 .

$$p_{X_1, X_2}(x_1, x_2) = P[X_1 = x_1, X_2 = x_2]$$

for all $(x_1, x_2) \in \mathcal{D}$.

Conditions for joint pmf

- $0 \leq p_{X_1, X_2}(x_1, x_2) \leq 1$.
- $\sum \sum_{(x_1, x_2) \in \mathcal{D}} p_{X_1, X_2}(x_1, x_2) = 1$.
- $P[(X, Y) \in A] = \sum \sum_{(x_1, x_2) \in A} p_{X_1, X_2}(x_1, x_2)$.

Supplementary Example 2.1-A

- Roll a pair of fair dice.
- Let X denote the smaller and Y the larger outcome on the dice.
- Find the joint pmf.

Note

- The support of a discrete random vector (X_1, X_2) is defined as

$$\{(x_1, x_2) : p_{X_1, X_2}(x_1, x_2) > 0\}.$$

- The definition of a pmf is extended to a convenient set by using zero elsewhere.

Continuous random vector

定義 3.3.

- A random vector (X_1, X_2) with space $\mathcal{D}$ is of the continuous type if its cdf $F_{X_1, X_2}(x_1, x_2)$ is continuous.

- $F_{X_1, X_2}(x_1, x_2)$ can be expressed as

$$F_{X_1, X_2}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{X_1, X_2}(w_1, w_2) dw_1 dw_2,$$

for all $(x_1, x_2) \in \mathcal{D}$.

- The integrand is called the joint probability density function (pdf) of (X_1, X_2) .

Note

- We have

$$\frac{\partial^2 F_{X_1, X_2}(x_1, x_2)}{\partial x_1 \partial x_2} = f_{X_1, X_2}(x_1, x_2)$$

except possibly on events that have probability zero.

Conditions for joint pdf

- $0 \leq f_{X_1, X_2}(x_1, x_2)$.
- $\int \int_{(x_1, x_2) \in \mathcal{D}} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 1$.
- For an event A ,

$$P[(X, Y) \in A] = \int \int_{(x_1, x_2) \in A} f_{X_1, X_2}(x_1, x_2) dx_1 dx_2.$$

Note

- The support of a continuous random vector (X_1, X_2) is defined as

$$\{(x_1, x_2) : f_{X_1, X_2}(x_1, x_2) > 0\}.$$

- The support is denoted as $\mathcal{S}$.
- The definition of a pdf is extended to $\mathbb{R}^2$ by using zero elsewhere.

Special event

- Let (X_1, X_2) be a random vector.
- Consider the event $\{X_1 \leq x_1\}$.
- We have

$$\{X_1 \leq x_1\} = \{X_1 \leq x_1\} \cap \{-\infty < X_2 < \infty\}$$

- The probability of this event is

$$F_{X_1}(x_1) = P[X_1 \leq x_1, -\infty < X_2 < \infty]$$

for all $x_1 \in \mathbb{R}$.

Marginal pmf

- Let (X_1, X_2) be a discrete random vector.
- Find the probability of $\{X_1 \leq x_1\}$ and the pmf of X_1 .

Supplementary Example 2.1-B

- Let the joint pmf of X and Y be defined by

$$f(x, y) = \frac{x + y}{21}, x = 1, 2, 3; y = 1, 2.$$

- Find $f_X(x)$ and $f_Y(y)$.

Marginal pdf

- Let (X_1, X_2) be a continuous random vector.
- Find the probability of $\{X_1 \leq x_1\}$ and the pdf of X_1 .

Example 2.1.4

- Let X and Y have the joint pdf

$$f(x, y) = \begin{cases} x + y & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}.$$

- Find the marginal pdf of X and Y , respectively.
- Find $P[X + Y \leq 1]$.

Expectation – Discrete type

- Let X_1 and X_2 be random variables of the discrete type with the joint pmf $f(x_1, x_2)$ on the space S .
- If $u(X_1, X_2)$ is a function of X_1 and X_2 , then

$$E[u(X_1, X_2)] = \sum_{(x_1, x_2) \in S} \sum u(x_1, x_2) f(x_1, x_2),$$

if it exists, is called the mathematical expectation of $u(X_1, X_2)$.

Remark

- If

$$\sum_{(x_1, x_2) \in S} \sum |u(x_1, x_2)| f(x_1, x_2),$$

converges, then $E[u(X_1, X_2)]$ exists.

Expectation – Continuous type

- Let X_1 and X_2 be random variables of the discrete type with the joint pdf $f(x_1, x_2)$ on the space S .
- If $u(X_1, X_2)$ is a function of X_1 and X_2 , then

$$E[u(X_1, X_2)] = \int \int_{(x_1, x_2) \in S} u(x_1, x_2) f(x_1, x_2) dx_1 dx_2,$$

if it exists, is called the mathematical expectation of $u(X_1, X_2)$.

Remark

- If

$$\int \int_{(x_1, x_2) \in S} |u(x_1, x_2)| f(x_1, x_2) dx_1 dx_2 < \infty,$$

converges, then $E[u(X_1, X_2)]$ exists.

Expectation

定理 3.1.

- Let (X_1, X_2) be a random vector.
- Let $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ be random variables whose expectations exist.
- For all real numbers k_1 and k_2 , we have

$$E[k_1 Y_1 + k_2 Y_2] = k_1 E[Y_1] + k_2 E[Y_2].$$

Moment generating function

定義 3.4.

- Let $X = (X_1, X_2)'$ be a random vector.
- If $E[e^{t_1 X_1 + t_2 X_2}]$ exists for $|t_1| < h_1$ and $|t_2| < h_2$, where h_1 and h_2 are positive.
- It is denoted by $M_{X_1, X_2}(t_1, t_2)$ and is called the moment generating function (mgf) of X .

Note

- Let $t = (t_1, t_2)'$.
- Then we can write the mgf of X as

$$M_{X_1, X_2}(t_1, t_2) = E[e^{t' X}].$$

- The mgfs of X_1 and X_2 are immediately seen to be $M_{X_1, X_2}(t_1, 0)$ and $M_{X_1, X_2}(0, t_2)$.

Example 2.1.7

- Let X and Y have the joint pdf

$$f(x, y) = \begin{cases} e^{-y} & 0 < x < y < \infty \\ 0 & \text{elsewhere} \end{cases}.$$

- Find $M(t_1, t_2)$, $M(t_1, 0)$ and $M(0, t_2)$.

Expected value**定義 3.5.**

- Let $X = (X_1, X_2)'$ be a random vector.
- Then the expected value of X exists if the expectations of X_1 and X_2 exist.
- If it exists, then the expected value is given by

$$E[X] = \begin{bmatrix} E[X_1] \\ E[X_2] \end{bmatrix}$$

3.4 Transformations: Bivariate random variables**Transformations – Discrete**

- Let $p_{X_1, X_2}(x_1, x_2)$ be the joint pmf of two discrete-type random variables X_1 and X_2 with the support $\mathcal{S}$.
- Let $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ define a one-to-one transformation that maps $\mathcal{S}$ onto $\mathcal{T}$.
- The joint pmf of the two new random variables $Y_1 = u_1(X_1, X_2)$ and $Y_2 = u_2(X_1, X_2)$ is given by

$$p_{Y_1, Y_2}(y_1, y_2) = \begin{cases} p_{X_1, X_2}(\omega_1(y_1, y_2), \omega_2(y_1, y_2)), & (y_1, y_2) \in \mathcal{T} \\ 0 & \text{elsewhere,} \end{cases}$$

where $x_1 = \omega_1(y_1, y_2)$, $x_2 = \omega_2(y_1, y_2)$ is the single-valued inverse of $y_1 = u_1(x_1, x_2)$, $y_2 = u_2(x_1, x_2)$.

Example 2.2.2

- Consider an experiment in which a person chooses at random a point (X_1, X_2) from the unit square $\mathcal{S} = \{(x_1, x_2) : 0 < x_1 < 1, 0 < x_2 < 1\}$.
- Suppose that our interest is not in X_1 or in X_2 , but in $Z = X_1 + X_2$.
- Find the pdf of Z .

Continuous transformation

- Let (X_1, X_2) have a jointly continuous distribution with pdf $f_{X_1, X_2}(x_1, x_2)$ and support set $\mathcal{S}$.
- Consider the transformed random vector $(Y_1, Y_2) = T(X_1, X_2)$ where T is a one-to-one continuous transformation.
- Let $\mathcal{T} = T(\mathcal{S})$ denote the support of (Y_1, Y_2) .
- Rewrite the transformation in terms of its components as

$$(Y_1, Y_2) = T(X_1, X_2) = (u_1(X_1, X_2), u_2(X_1, X_2)),$$

where the functions $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ define T .

Continuous transformation

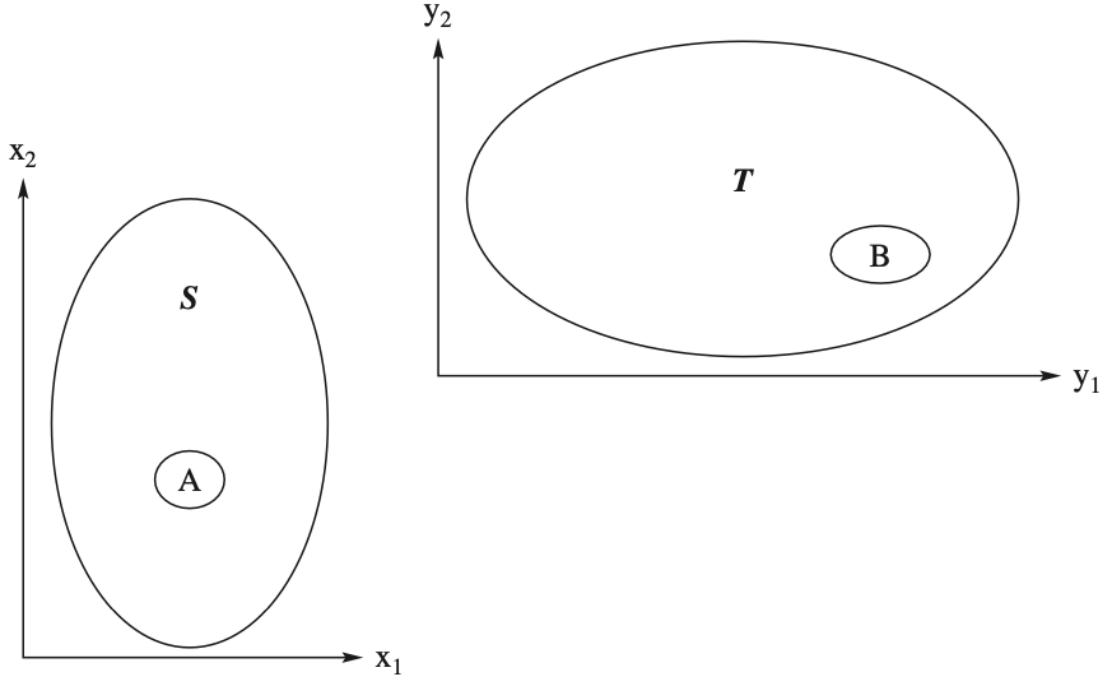
- Since the transformation is one-to-one, the inverse transformation T^{-1} exists.
- Write $x_1 = w_1(y_1, y_2)$, $x_2 = w_2(y_1, y_2)$.
- Define the Jacobian of the transformation which is the determinant of order 2 given by

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

Continuous transformation

- Let B be any region in $\mathcal{T}$.
- Let $A = T^{-1}(B)$.
- Because the transformation T is one-to-one,

$$\begin{aligned} P[(X_1, X_2) \in A] &= P[T(X_1, X_2) \in T(A)] \\ &= P[(Y_1, Y_2) \in B]. \end{aligned}$$



Continuous transformation

- Based on the change-in-variable technique, we have

$$\begin{aligned}
 P[(X_1, X_2) \in A] &= \int \int_A f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 \\
 &= \int \int_{T(A)} f_{X_1, X_2}(T^{-1}(y_1, y_2)) |J| dy_1 dy_2 \\
 &= \int \int_B f_{X_1, X_2}(w_1(y_1, y_2), w_2(y_1, y_2)) |J| dy_1 dy_2
 \end{aligned}$$

- The pdf of (Y_1, Y_2) is

$$f_{Y_1, Y_2}(y_1, y_2) = \begin{cases} f_{X_1, X_2}(w_1(y_1, y_2), w_2(y_1, y_2)) |J|, & (y_1, y_2) \in \mathcal{T} \\ 0 & \text{elsewhere} \end{cases}$$

Example 2.2.3

- Consider an experiment in which a person chooses at random a point (X_1, X_2) from the unit square $\mathcal{S} = \{(x_1, x_2) : 0 < x_1 < 1, 0 < x_2 < 1\}$.
- Find the joint pdf of $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$.

Example 2.2.5

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 10x_1x_2^2, & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find the pdf of $Y_1 = X_1/X_2$.

3.5 Conditional Distributions and Expectations**Conditional distribution for discrete r.v.**

- Let X_1 and X_2 denote random variables of the discrete type.
- Let the joint pmf be denoted as $p_{X_1, X_2}(x_1, x_2)$ that is positive on the support set $\mathcal{S}$ and is zero elsewhere.
- Let $p_{X_1}(x_1)$ and $p_{X_2}(x_2)$ denote, respectively, the marginal probability mass functions of X_1 and X_2 .
- Let x_1 be a point in the support of X_1 , i.e., $p_{X_1}(x_1) > 0$.
- Using the definition of conditional probability, we have

$$\begin{aligned} p_{X_2|X_1}(x_2 | x_1) &= P[X_2 = x_2 | X_1 = x_1] \\ &= \frac{P[X_1 = x_1, X_2 = x_2]}{P[X_1 = x_1]} \\ &= \frac{p_{X_1, X_2}(x_1, x_2)}{p_{X_1}(x_1)} \end{aligned}$$

for all $x_2 \in \mathcal{S}_{X_2}$.

Properties

- $\sum_{x_2} p_{X_2|X_1}(x_2|x_1) = 1$.
- $p_{X_2|X_1}(x_2|x_1) > 0$ for all $x_2 \in \mathcal{S}_{X_2}$.
- $p_{X_2|X_1}(x_2|x_1)$ is called the conditional pmf of the discrete type of random variable.

Conditional distribution for continuous r.v.

- Let X_1 and X_2 denote random variables of the continuous type.
- Let the joint pdf be denoted as $f_{X_1, X_2}(x_1, x_2)$ that is positive on the support set $\mathcal{S}$ and is zero elsewhere.
- Let $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ denote, respectively, the marginal probability density functions of X_1 and X_2 .

- When $f_{X_1}(x_1) > 0$, we define

$$f_{X_2|X_1}(x_2 | x_1) = \frac{f_{X_1, X_2}(x_1, x_2)}{f_{X_1}(x_1)}.$$

Properties

- $\int_{-\infty}^{\infty} f_{X_2|X_1}(x_2|x_1)dx_2 = 1$.
- $f_{X_2|X_1}(x_2 | x_1) \geq 0$ for all $x_2 \in \mathcal{S}_{X_2}$.
- $f_{X_2|X_1}(x_2|x_1)$ is called the conditional pdf of the continuous type of random variable.

Conditional probability of continuous r.v.

- If the random variables are of the continuous type, the conditional probability that $a < X_2 < b$, given that $X_1 = x_1$, is

$$P[a < X_2 < b | X_1 = x_1] = \int_a^b f_{X_2|X_1}(x_2|x_1)dx_2$$

Conditional expectation of r.v.

- If $u(X_2)$ is a function of X_2 , the conditional expectation of $u(X_2)$, given that $X_1 = x_1$, if it exists, is given by

$$E[u(X_2)|X_1 = x_1] = \begin{cases} \sum_{x_2} u(x_2)p_{X_2|X_1}(x_2|x_1), & \text{Discrete} \\ \int_{x_2} u(x_2)f_{X_2|X_1}(x_2|x_1)dx_2, & \text{Continuous} \end{cases}$$

- The conditional variance of X_2 , given that $X_1 = x_1$, if it exists, is given by

$$\begin{aligned} \text{Var}(X_2 | X_1 = x_1) &= E[(X_2 - E[X_2 | X_1 = x_1])^2 | X_1 = x_1] \\ &= E[X_2^2 | X_1 = x_1] - (E[X_2 | X_1 = x_1])^2 \end{aligned}$$

Example 2.3.1

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find $f_{X_1|X_2}(x_1|x_2)$ and $f_{X_2|X_1}(x_2|x_1)$.

Example 2.3.1 (continued)

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find $E[X_1|x_2]$ and $\text{Var}(X_1|x_2)$.

Supplementary Example 2.3-A

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find $P[0 < X_1 < \frac{1}{2} | X_2 = \frac{3}{4}]$ and $P[0 < X_1 < \frac{1}{2}]$

Conditional variance and variance

定理 3.2.

- Let (X_1, X_2) be a random vector such that the variance of X_2 is finite.
- We have
 - $E[E[X_2|X_1]] = E[X_2]$.
 - $\text{Var}(E[X_2|X_1]) \leq \text{Var}(X_2)$.

3.6 Independent random variables**Independence**

定義 3.6.

- Let the random variables X_1 and X_2 have the joint pdf $f(x_1, x_2)$ [joint pmf $p(x_1, x_2)$].
- Let the marginal pdfs [pmfs] $f_1(x_1)[p_1(x_1)]$ and $f_2(x_2)[p_2(x_2)]$.
- X_1 and X_2 are said to be independent if, and only if,

$$f(x_1, x_2) \equiv f_1(x_1)f_2(x_2) [p(x_1, x_2) \equiv p_1(x_1)p_2(x_2)].$$

- Random variables that are not independent are said to be dependent.

Remark

- The product of two positive functions $f_1(x_1)f_2(x_2)$ means a function that is positive on the product space.
- There may be certain points $(x_1, x_2) \in \mathcal{S}$ at which $f(x_1, x_2) \neq f_1(x_1)f_2(x_2)$. However, if A is the set of points (x_1, x_2) at which the equality does not hold, then $P(A) = 0$.

Supplementary Example 2.5-A

- Suppose an urn contains 10 blue, 8 red, and 7 yellow balls that are the same except for color.
- Suppose 4 balls are drawn without replacement.
- Let X and Y be the number of red and blue balls drawn.
- Are X and Y independent?

Independence

定理 3.3.

- Let the random variables X_1 and X_2 have supports $\mathcal{S}_1$ and $\mathcal{S}_2$, respectively, and have the joint pdf $f(x_1, x_2)$.
- Then X_1 and X_2 are independent if and only if $f(x_1, x_2)$ can be written as a product of a nonnegative function of x_1 and a nonnegative function of x_2 .
- That is,

$$f(x_1, x_2) \equiv g(x_1)h(x_2),$$

where $g(x_1) > 0$, $x_1 \in \mathcal{S}_1$, zero elsewhere, and $h(x_2) > 0$, $x_2 \in \mathcal{S}_2$, zero elsewhere.

Class Example 2.5-A

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 4xe^{-x^2}ye^{-y^2}, & x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

- Are X_1 and X_2 independent?

Example 2.5.2

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 8x_1x_2 & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Are X and Y independent?

Independence

定理 3.4.

- Let X_1 and X_2 have the joint cdf $F(x_1, x_2)$.
- Let X_1 and X_2 have the marginal cdfs $F_1(x_1)$ and $F_2(x_2)$.

- Then X_1 and X_2 are independent if and only if

$$F(x_1, x_2) = F_1(x_1)F_2(x_2)$$

for all $(x_1, x_2) \in R^2$.

Independence

定理 3.5.

- X_1 and X_2 are independent if and only if

$$P[a < X_1 \leq b, c < X_2 \leq d] = P[a < X_1 \leq b]P[c < X_2 \leq d]$$

for every $a < b$ and $c < d$, where a, b, c, d are constants.

Independence

定理 3.6.

- Suppose X_1 and X_2 are independent.
- Suppose $E[u(X_1)] < \infty$ and $E[\nu(X_2)] < \infty$.
- We have

$$E[u(X_1)\nu(X_2)] = E[u(X_1)]E[\nu(X_2)].$$

Independence

定理 3.7.

- Suppose $M(t_1, t_2) < \infty$.
- Then X_1 and X_2 are independent if and only if

$$M(t_1, t_2) = M(t_1, 0)M(0, t_2).$$

Example 2.5.5

- Let X_1 and X_2 have the joint pdf

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} e^{-x_2} & 0 < x_1 < x_2 < \infty \\ 0 & \text{elsewhere} \end{cases}$$

- Find $M(t_1, t_2)$.
- Show that $M(t_1, t_2) \neq M(t_1, 0)M(0, t_2)$.

3.7 The correlation coefficient

Covariance

定義 3.7.

- Let (X, Y) have a joint distribution.
- Denote the means of X and Y by μ_1 and μ_2 and their respective variances by σ_1^2 and σ_2^2 .
- The covariance of (X, Y) is denoted by $\text{cov}(X, Y)$ and is defined by the expectation

$$\text{cov}(X, Y) = E[(X - \mu_1)(Y - \mu_2)].$$

Note

- The covariance of X and Y can also be expressed as

$$\text{cov}(X, Y) = E[XY] - \mu_1\mu_2.$$

Correlation coefficient

定義 3.8.

- If each of σ_1 and σ_2 is positive, then the correlation coefficient between X and Y is defined by

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_1\sigma_2}.$$

Note

- The expected value of the product of two random variables is equal to the product of their expectations plus their covariance;

$$E[XY] = \mu_1\mu_2 + \text{Cov}(X, Y) = \mu_1\mu_2 + \rho\sigma_1\sigma_2.$$

Supplementary Example 2.4-A

- A fair coin is flipped three times.
- X_1 is the number of heads on the first two flips while X_2 is the number of heads on all three flips.
- Find the correlation coefficient of (X_1, X_2) .

Correlation coefficient¹

定理 3.8.

- For all jointly distributed random variables (X, Y) whose correlation coefficient ρ exists, $-1 \leq \rho \leq 1$.

¹Hint: Define $h(v) = E\{[(X - \mu_1) + v(Y - \mu_2)]^2\}$

Note

- A quadratic equation is said positive when its discriminant is negative.
- Define the quadratic equation as $y = ax^2 + bx + c$.
- The discriminant is $b^2 - 4ac$.

Independence

定理 3.9.

- If X and Y are independent random variables, then

$$\rho = 0.$$

Class Example 2.4-A

- Let X and Y be jointly discrete random variables whose distribution has mass 1/4 at each of the four points $(-1, 0)$, $(0, -1)$, $(1, 0)$ and $(0, 1)$.
- Find the correlation coefficient of (X_1, X_2) .
- Are X and Y independent?

Linear function

定理 3.10.

- Suppose (X, Y) have a joint distribution with the variances of X and Y finite and positive.
- Denote the means and variances of X and Y by μ_1 , μ_2 and σ_1^2 , σ_2^2 .
- Let ρ be the correlation coefficient between X and Y .
- If $E[Y|X]$ is linear in X then

$$E[Y|X] = \mu_2 + \rho \frac{\sigma_2}{\sigma_1} (X - \mu_1)$$

and

$$E[\text{Var}(Y|X)] = \sigma_2^2(1 - \rho^2).$$

Example 2.4.2

- Let the random variables X and Y have the linear conditional means $E(Y|x) = 4x + 3$ and $E(X|y) = \frac{1}{16}y - 3$.
- Can μ_1 , μ_2 , ρ , σ_1 and σ_2 be obtained?

Stage 2 — Deepen：由二維推廣到多維

從兩個變數的逐項計算，提升為 **random vector**、**matrix** 與一般維度的表示。

大二教材主要集中在雙變量情形；本階段是本章的主要新內容。學習重點是用向量與矩陣統整平均數、變異數與共變異數，理解 **covariance matrix** 的結構，並將二維 **transformation** 的方法推廣到多維與 many-to-one 情形。

3.8 Extension to several random variables

Random vector

定義 3.9.

- Consider a random experiment with a sample space $\mathcal{C}$.
- Let the random variable X_i assign to each element $c \in \mathcal{C}$ one and only one real number $X_i(c) = x_i, i = 1, 2, \dots, n$.
- Then,
 - $X = (X_1, X_2, \dots, X_n)$ is a n -dimensional random vector.
 - The space of X is the set of ordered n -tuples

$$\mathcal{D} = \{(x_1, x_2, \dots, x_n) : x_1 = X_1(c), \dots, x_n = X_n(c), c \in \mathcal{C}\}.$$

- Furthermore, let A be a subset of the space $\mathcal{D}$. Then $P[(X_1, \dots, X_n) \in A] = P[C]$, where $C = \{c : c \in \mathcal{C} \text{ and } (X_1(c), X_2(c), \dots, X_n(c)) \in A\}$.

Cumulative distribution function (cdf)

- Let $A \subset \mathcal{D}$.
- Denote the probability of the event A as $P_{X_1, X_2, \dots, X_n}[A]$.
- $P_{X_1, X_2, \dots, X_n}[A]$ is uniquely determined by the cumulative distribution function (cdf), which is given by

$$F_X(x_1, x_2, \dots, x_n) = P[X_1 \leq x_1, \dots, X_n \leq x_n],$$

for all $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$.

Cumulative distribution function (cdf)

- Discrete random vector

$$F_X(x_1, x_2, \dots, x_n) = \sum \cdots \sum_{\omega_1 \leq x_1, \dots, \omega_n \leq x_n} p(\omega_1, \dots, \omega_n)$$

- Continuous random vector

$$F_X(x_1, x_2, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(\omega_1, \dots, \omega_n) d\omega_1 \cdots d\omega_n.$$

Moreover,

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_X(x) = f(x).$$

Example 2.6.1

- Let the random variables X, Y and Z have the joint pdf defined as

$$f(x, y, z) = \begin{cases} e^{-x+y+z} & 0 < x, y, z < \infty \\ 0 & \text{elsewhere} \end{cases}$$

- Find the distribution function of X, Y and Z .

Expectation for discrete random vector

- Let $X = (X_1, X_2, \dots, X_n)$ be a random vector.
- Let $Y = u(X_1, X_2, \dots, X_n)$ for some function u .
- Then

$$E[u(X_1, X_2, \dots, X_n)] = \sum \cdots \sum_{(x_1, \dots, x_n) \in \mathcal{D}} u(x_1, \dots, x_n) f(x_1, \dots, x_n),$$

if it exists, is called the mathematical expectation of $u(X_1, \dots, X_n)$.

Expectation for continuous random vector

- Let $X = (X_1, X_2, \dots, X_n)$ be a random vector.
- Let $Y = u(X_1, X_2, \dots, X_n)$ for some function u .
- Then

$$E[u(X_1, \dots, X_n)] = \int \cdots \int_{(x_1, \dots, x_n) \in \mathcal{D}} u(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_1 \cdots dx_n,$$

if it exists, is called the mathematical expectation of $u(X_1, \dots, X_n)$.

Marginal pmf

- Let $(X_1, \dots, X_n)$ be a random vector.
- Find the pmf of X_1 .

Note

- Take any group of $k < n$ of these random variables and find the joint pmf of them.
- This joint pmf is called the marginal pmf of this particular group of k variables.

Marginal pdf

- Let $(X_1, \dots, X_n)$ be a random vector.
- Find the probability of $\{X_1 \leq x_1\}$ and the pdf of X_1 .

Note

- Take any group of $k < n$ of these random variables and find the joint pdf of them.
- This joint pdf is called the marginal pdf of this particular group of k variables.

Conditional distribution for discrete type

- Let x_1 be a point in the support of X_1 , i.e., $p_{X_1}(x_1) > 0$.
- Using the definition of conditional probability, we have

$$\begin{aligned} p_{X_2, \dots, X_n | X_1}(x_2, \dots, x_n | x_1) &= P[X_2 = x_2, \dots, X_n = x_n | X_1 = x_1] \\ &= \frac{p_X(x_1, x_2, \dots, x_n)}{p_{X_1}(x_1)}. \end{aligned}$$

Conditional distribution for continuous type

- When $f_{X_1}(x_1) > 0$, we define

$$f_{X_2, \dots, X_n | X_1}(x_2, \dots, x_n | x_1) = \frac{f_X(x_1, x_2, \dots, x_n)}{f_{X_1}(x_1)}.$$

- The joint conditional pdf of any $n - 1$ random variables, say $X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n$, given $X_i = x_i$, is defined as the joint pdf of $X_1, \dots, X_n$ divided by the marginal pdf $f_{X_i}(x_i)$, provided that $f_{X_i}(x_i) > 0$.

Conditional expectation of a discrete random vector

- If $u(X_2, \dots, X_n)$ is a function of $X_2, \dots, X_n$, the conditional expectation of $u(X_2, \dots, X_n)$, given that $X_1 = x_1$, if it exists, is given by

$$E[u(X_2, \dots, X_n) | X_1 = x_1] = \sum \cdots \sum_{x_2, \dots, x_n} u(x_2, \dots, x_n) p_{X_2, \dots, X_n | X_1}(x_2, \dots, x_n | x_1),$$

Conditional expectation of a continuous random vector

- If $u(X_2, \dots, X_n)$ is a function of $X_2, \dots, X_n$, the conditional expectation of $u(X_2, \dots, X_n)$, given that $X_1 = x_1$, if it exists, is given by

$$E[u(X_2, \dots, X_n) | X_1 = x_1] = \int \cdots \int_{\mathcal{D}_{2:n}} u(x_2, \dots, x_n) f_{X_2, \dots, X_n | X_1}(x_2, \dots, x_n | x_1) dx_2 \cdots dx_n,$$

Independence

定義 3.10.

- Let the random variables $X_1, \dots, X_n$ have the joint pdf $f(x_1, \dots, x_n)$ [joint pmf $p(x_1, \dots, x_n)$].
- Let the marginal pdfs [pmfs] $f_1(x_1)[p_1(x_1)], \dots, f_n(x_n)[p_n(x_n)]$.
- $X_1, \dots, X_n$ are said to be mutually independent if, and only if,

$$f(x_1, \dots, x_n) \equiv f_1(x_1) \cdots f_n(x_n) [p(x_1, \dots, x_n) \equiv p_1(x_1) \cdots p_n(x_n)].$$

- Random variables that are not independent are said to be dependent.

Independence

定理 3.11.

- Suppose $X_1, \dots, X_n$ are independent.
- Suppose $E[u(X_i)] < \infty, i = 1, 2, \dots, n$.
- We have

$$E[u(X_1) \cdots u(X_n)] = E[u(X_1)] \cdots E[u(X_n)] = \prod_{i=1}^n E[u(X_i)]$$

Moment generating function

定義 3.11.

- Let $X = (X_1, X_2, \dots, X_n)$ be a random vector.
- If $E[e^{\sum_{i=1}^n t_i X_i}]$ exists for $|t_i| < h_i, i = 1, \dots, n$ where $h_i > 0, i = 1, \dots, n$.
- It is denoted by $M_X(t_1, \dots, t_n)$ and is called the moment generating function (mgf) of X .

Independence

定理 3.12.

- Suppose $X_1, X_2, \dots, X_n$ are n mutually independent random variables.
- Suppose, for all $i = 1, 2, \dots, n$, X_i has mgf $M_i(t)$, for $|t| < h_i$, where $h_i > 0$.
- Let $T = \sum_{i=1}^n k_i X_i$, where $k_1, k_2, \dots, k_n$ are constants.
- Then T has the mgf given by

$$M_T(t) = \prod_{i=1}^n M_i(k_i t),$$

where $-\min_i(h_i) < t < \min_i h_i$.

Supplementary Example 2.6-A

- Let the random variables X_1 , X_2 and X_3 be three mutually independent random variable and let each have the pdf

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find the distribution function and pdf of $Y = \max_i X_i$.

Note

- Unless there is a possible misunderstanding between mutual and pairwise independence, we usually drop the modifier mutual.
- If several random variables are mutually independent and have the same distribution, we say that they are independent and identically distributed, which we abbreviate as iid.

Independence

- Suppose $X_1, X_2, \dots, X_n$ are iid random variable with the common mgf $M(t)$, for $|t| < h$, where $h > 0$.
- Let $T = \sum_{i=1}^n X_i$.
- Then T has the mgf given by

$$M_T(t) = [M(t)]^n,$$

where $|t| < h$.

Random matrix

- Let $X = (X_1, \dots, X_n)'$ be an n -dimensional random vector.
- Define $E(X) = (E[X_1], \dots, E[X_n])'$.
- Suppose W is an $m \times n$ matrix of random variables with the ij th component W_{ij} , $1 \leq i \leq m$ and $1 \leq j \leq n$.
- Define the expectation of a random matrix

$$E[W] = [E(W_{ij})].$$

Random matrix

定理 3.13.

- Let W_1 and W_2 be $m \times n$ matrices of random variables.
- Let A_1 and A_2 be $k \times m$ matrices of constants.
- Let B be an $n \times l$ matrix of constants.
- Then, we have

- $E[A_1W_1 + A_2W_2] = A_1E[W_1] + A_2E[W_2]$.
- $E[A_1W_1B] = A_1E[W_1]B$

Random vector

- Let $X = (X_1, \dots, X_n)'$ be an n -dimensional random vector, such that $\sigma_i^2 = \sigma_{ii} = \text{Var}(X_i) < \infty$.
- The mean of X is $\mu = E[X]$. and we define its variance-covariance matrix as

$$\text{Cov}(X) = E[(X - \mu)(X - \mu)'] = [\sigma_{ij}]$$

whose the (i, j) th off diagonal entry is $\text{Cov}(X_i, X_j)$ and the (i, i) th diagonal entry is $\sigma_i^2 = \text{Var}(X_i)$.

Example 2.6.3

- Let the random variables X and Y have the joint pdf of X and Y is

$$f_{X,Y}(x, y) = \begin{cases} e^{-y} & 0 < x < y < \infty \\ 0 & \text{elsewhere} \end{cases}$$

- Let $Z = (X, Y)'$.
- Find $E[Z]$ and $\text{Cov}(Z)$.

Random vector

定理 3.14.

- Let $X = (X_1, \dots, X_n)'$ be an n -dimensional random vector, such that $\sigma_i^2 = \sigma_{ii} = \text{Var}(X_i) < \infty$.
- Let A be an $m \times n$ matrices of constants.
- Then, we have
 - $\text{Cov}(X) = E[XX'] - \mu\mu'$
 - $\text{Cov}(AX) = A \text{Cov}(X) A'$.

Note

- All variance-covariance matrices are positive semidefinite; that is,

$$a' \text{Cov}(X) a \geq 0,$$

for all vectors $a \in \mathbb{R}^n$.

- Let X be a random vector.

- Let a be any $n \times 1$ vector of constants.
- Then $Y = a'X$ is a random variable and, hence, has nonnegative variance; i.e.,

$$0 \leq \text{Var}(Y) = \text{Var}(a'X) = a' \text{Cov}(X)a.$$

3.9 Transformations for several random variables

Continuous transformation

- Let $(X_1, \dots, X_n)$ have a jointly continuous distribution with pdf $f(x_1, \dots, x_n)$ and support set $\mathcal{S}$.
- Consider a one-to-one transformation that maps $\mathcal{S}$ onto $\mathcal{T}$.
- Let $y_1 = u_1(x_1, x_2, \dots, x_n)$, $y_2 = u_2(x_1, x_2, \dots, x_n)$, $\dots$ and $y_n = u_n(x_1, x_2, \dots, x_n)$.
- Let the inverse function be denoted as

$$\begin{aligned} x_1 &= w_1(y_1, y_2, \dots, y_n) \\ x_2 &= w_2(y_1, y_2, \dots, y_n) \\ &\vdots \\ x_n &= w_n(y_1, y_2, \dots, y_n) \end{aligned}$$

Continuous transformation

- Define the Jacobian of the transformation as

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \dots & \frac{\partial x_1}{\partial y_n} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \dots & \frac{\partial x_2}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \frac{\partial x_n}{\partial y_2} & \dots & \frac{\partial x_n}{\partial y_n} \end{vmatrix}$$

- J can not be identically zero in $\mathcal{T}$.

Continuous transformation

- Based on the change-in-variable technique, we have

$$\int \dots \int_A f(x_1, x_2, \dots, x_n) dx_1 \dots dx_n = \int \dots \int_B f(w_1(y), \dots, w_n(y)) |J| dy_1 \dots dy_n$$

- The pdf of $(Y_1, Y_2, \dots, Y_n)$ is

$$f_Y(y_1, y_2, \dots, y_n) = \begin{cases} f(w_1(y_1, y_2, \dots, y_n), \dots, w_n(y_1, y_2, \dots, y_n)) |J|, & y \in \mathcal{T} \\ 0 & \text{elsewhere} \end{cases}$$

Example 2.7.1

- Let X_1, X_2 and X_3 have the joint pdf

$$f(x_1, x_2, x_3) = \begin{cases} 48x_1x_2x_3 & 0 < x_1 < x_2 < x_3 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Find the joint pdf of Y_1, Y_2 and Y_3 .
- Find the marginal pdf of Y_1 and that of Y_2 .

Class Example 2.7-A — Many to one

- Let X have the Cauchy pdf

$$f(x) = \begin{cases} \frac{1}{\pi(1+x^2)}, & -\infty < x < \infty \\ 0 & \text{elsewhere} \end{cases}$$

- Let $Y = X^2$.
- Find the pdf of Y .

Continuous transformation

- Let $(X_1, \dots, X_n)$ have a jointly continuous distribution with pdf $f(x_1, \dots, x_n)$ and support set $\mathcal{S}$.
- Consider the transformation $y_1 = u_1(x_1, x_2, \dots, x_n), y_2 = u_2(x_1, x_2, \dots, x_n), \dots$ and $y_n = u_n(x_1, x_2, \dots, x_n)$, which maps $\mathcal{S}$ to $\mathcal{T}$.
- To each point of $\mathcal{S}$ there corresponds, of course, only one point in $\mathcal{T}$; but to a point in $\mathcal{T}$ there may correspond more than one point in $\mathcal{S}$.

Continuous transformation

- Suppose that we can represent $\mathcal{S}$ as the union of a finite number, say k , of mutually disjoint sets $A_1, A_2, \dots, A_k$ so that

$$y_1 = u_1(x_1, x_2, \dots, x_n), \dots, y_n = u_n(x_1, x_2, \dots, x_n)$$

define a one-to-one transformation of each A_i onto $\mathcal{T}$.

- To each point in $\mathcal{T}$, there corresponds exactly one point in each of $A_1, A_2, \dots, A_k$.
- For $i = 1, \dots, k$, let the inverse function be denoted as

$$\begin{aligned} x_1 &= w_{1i}(y_1, y_2, \dots, y_n) \\ x_2 &= w_{2i}(y_1, y_2, \dots, y_n) \\ &\vdots \\ x_n &= w_{ni}(y_1, y_2, \dots, y_n) \end{aligned}$$

Continuous transformation

- Define the Jacobian of the transformation as

$$J_i = \begin{vmatrix} \frac{\partial x_{1i}}{\partial y_1} & \frac{\partial x_{1i}}{\partial y_2} & \cdots & \frac{\partial x_{1i}}{\partial y_n} \\ \frac{\partial x_{2i}}{\partial y_1} & \frac{\partial x_{2i}}{\partial y_2} & \cdots & \frac{\partial x_{2i}}{\partial y_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_{ni}}{\partial y_1} & \frac{\partial x_{ni}}{\partial y_2} & \cdots & \frac{\partial x_{ni}}{\partial y_n} \end{vmatrix}, i = 1, \dots, k.$$

- $J_i, i = 1, \dots, k$, can not be identically zero in $\mathcal{T}$.
- The pdf of $(Y_1, Y_2, \dots, Y_n)$ is

$$f_Y(y_1, y_2, \dots, y_n) = \sum_{i=1}^k f(w_{1i}(y_1, y_2, \dots, y_n), \dots, w_{ni}(y_1, y_2, \dots, y_n)) |J_i|$$

provided $y \in \mathcal{T}$.

Example 2.7.5

- Let X_1, X_2, X_3, X_4 be independent random variables with common pdf

$$f(x) = \begin{cases} e^{-x} & 0 < x \\ 0 & \text{elsewhere} \end{cases}$$

- Find the distribution function of $Y = \sum_{i=1}^4 X_i$ using the mgf technique.

Stage 3 — Connect : 連接抽樣分配與統計推論

把 **random vector** 的工具放回樣本 $X_1, \dots, X_n$ ，開始研究由樣本形成的統計量。

本階段以線性組合與 random sample 收束本章。學生應能計算線性統計量的期望、變異數與共變異數，辨認 $\bar{X}$ 與 S^2 是樣本的函數，並理解 $E(S^2) = \sigma^2$ 為何是後續估計理論的重要起點。

3.10 Linear combinations random variables

Linear combination

- Let $X = (X_1, \dots, X_n)'$ and $Y = (Y_1, \dots, Y_m)'$ denote two random vectors.
- Consider two linear combinations of these variables as

$$T = \sum_{i=1}^n a_i X_i$$

$$W = \sum_{i=1}^m b_i Y_i$$

for specified constants a_i and $b_i, i = 1, 2, \dots, n$.

- Suppose $E[X_i] = \mu_{X_i}, i = 1, 2, \dots, n$ and $E[Y_i] = \mu_{Y_i}, i = 1, 2, \dots, m$.

Mean and variance of T

定理 3.15.

- $E[T] = \sum_{i=1}^n a_i \mu_{X_i}$ and $E[W] = \sum_{i=1}^n b_i \mu_{Y_i}$.
- $\text{Var}(T) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j)$.
- $\text{Cov}(T, W) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{Cov}(X_i, Y_j)$.

Random sample

定義 3.12.

- If the random variables $X_1, X_2, \dots, X_n$ are independent and identically distributed.
- Each X_i has the same distribution.
- These random variables constitute a random sample of size n from that common distribution.
- Abbreviate independent and identically distributed by iid.

Example 2.8.2 — Sample Variance

- Let $X_i, i = 1, 2, \dots, n$, be a random sample with mean μ and variance σ^2 .
- Define the sample variance as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- Show $E[S^2] = \sigma^2$.

3.11 本章摘要

本章把單變量分配推廣到隨機向量，核心工具是邊際化、條件化、Jacobian 與共變異結構。

3.12 常見錯誤

- 將零相關直接解讀為獨立
- Jacobian 使用方向錯誤
- 只求聯合密度而未確認轉換後支持集
- 忽略條件分配中的正規化常數

3.13 章末練習

1. 從本章選出三個關鍵定義，分別寫出數學形式、成立條件及一個例子。
2. 選擇一個主要定理或方法，重做推導並標記每一步使用的假設。

3. 從原講義例題挑選一題，完整寫出解題目標、方法選擇、計算與結果解釋。
4. 設計一個會違反本章方法條件的反例，說明錯誤會出現在哪一步。

i 待逐章補強

建議後續為本章補入：中文概念導讀、完整解答、課堂練習難度標記，以及與下一章的銜接說明。